

Comparative Analysis of Classification Models for Type 2 Diabetes Prediction Using CDC BRFSS Survey Data

Introduction & Summary:

Diabetes is a blight on public health across the United States, and increasingly more so in developing countries as time progresses. Diabetes symptoms are debilitating and can impart significant financial strain on both individuals and the community. The advent of effective and available GLP-1 medications for the treatment of type 2 diabetes makes early detection and intervention more crucial than ever. This presents itself as a very compelling area of public health research, and we intend to analyze the CDC's Behavioral Risk Factor Surveillance System survey dataset with classification models.

Anticipated Objective:

The objective is threefold; build and compare a suite of classification models to predict type 2 diabetes diagnosis, evaluate model performance across both binary and multi-class output targets, and identify which models perform best with respect to interpretability, performance, and generalization.

Proposed Tasks:

Data Preparation: The dataset itself will require a large degree of EDA, preprocessing (including encoding, scaling, missing data, and imbalance) as well as feature selection/dimensionality reduction.

Model Training: The analysis will be testing a variety of classification models, including logistic regression, kernel methods, SVM, and neural network classifiers. The target classes will include both diabetic/non-diabetic as well as diabetic/pre-diabetic/non-diabetic, so we will explore both binary and multi-class model types.

Model Evaluation: The models will be evaluated using standardized metrics (accuracy, precision, recall, F1, AUC-ROC, and confusion matrices) with cross-validation. We will also compare the efficacy of the models when trained to predict a binary vs multi-class outcome. Special attention will be given to weighting false-positives vs false-negatives, as these have different relative disutility.

Report: We will write the report in the IEEE formatting using LaTeX. All code will be managed through a GitHub repository.

Schedule:

Week	Dates	Task(s)
1	3/30 – 4/5	EDA & Preprocessing
2	4/6 – 4/12	Logistic Regression Analysis
3	4/13 – 4/19	Kernel Methods & SVM Analysis
4	4/20 – 4/26	Neural Network Analysis
5	4/27 – 5/3	Model Comparison, Metric Collection, and Visualizations
6	5/4 – 5/11	Finalize Written Report

References:

1. CDC BRFSS Survey Data
2. Prior Diabetic Detection Models (for guidance on model direction)